

Gamification as a methodology to enhance analytical and sustainable engagement on social media

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Abstract

This research examines the impact of gamification on the development of critical thinking skills to counter misinformation, focusing specifically on students' roles as developers of educational software. Involving 107 university students, this study employs an experimental pretest–posttest methodology. An educational Escape Room was implemented, and its influence on critical and digital competencies was scrutinized. The procedure comprised three stages: pretest measurement, formative intervention, and posttest evaluation. A comparative analysis of the scores for the questions highlights a pressing need to enhance media and digital literacy. The findings reveal a predominance of responses indicating improvements in source verification and fact-checking. However, no significant differences were observed in the pre and post responses ($0.607 > 0.05$), leading to the conclusion that the implementation of a technological training program does not significantly contribute to acquiring skills for identifying falsehoods and deceptions on the internet. In conclusion, the study underscores the necessity to develop effective strategies to address misinformation and recognizes gamification as a motivating educational tool.

Keywords Digital literacy · Gamification · Teacher competencies · Equal education · Sustainable development

1 Introduction

Education and training in technology play a pivotal role in equipping citizens with the critical thinking abilities required to exercise discernment in the digital environment. This endeavour is situated within the realm of digital literacy, crucial not only for individuals to feel secure but also to critically engage with the information they encounter and consume. A deficiency in digital literacy gives rise to a digital divide, a disparity between those who have access to and proficiency in digital technology and those who do not. In accordance with the findings of Tomczyk et al. [33] in their investigation of the impediments to digital inclusion among the elderly, it is evident that a deficiency in digital skills can result in considerable exclusion within the information society. Tomczyk et al. [33] further assert that the phenomenon of digital exclusion, which has become increasingly conspicuous over the past three decades, epitomizes a distinct variant of social exclusion. Álvarez et al. [1] coined the term “cognitive digital divide” to describe the set of knowledge and skills necessary

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for the efficacious use of technology and information. Indicators of digital inclusion based on knowledge levels pertain to cognitive skills for handling technology securely and critically [32].

In times of crisis, managing disinformation emerges as one of the challenges we confront. Disinformation can wield significant influence, destabilising economies, fostering uncertainty, and swaying public opinion. This is mirrored in the findings of the European Union's Eurobarometer, wherein seven out of ten Europeans assert they frequently encounter online information they deem deceptive or distorting of the truth. Survey results from the European Union's Digital Education Action Plan 2021–2027 [8] highlight, as exemplars of competency, the abilities to discern between pertinent information and false data, manage vast quantities of information, and navigate the web safely.

Researchers investigating the consequences of misinformation agree on the necessity to formulate robust counter-measures [31]. Some of these approaches might rely on technologies designed to mitigate the effects of misleading information and strengthen critical assessment [28], ensuring that the credibility of science is not eroded, especially when portrayed as an institution unjustly exerting political influence [5].

1.1 The European digital competence framework for citizens (DIGCOMP)

European Institutions have acknowledged the significance of Digital Competencies for engagement in the Knowledge Society, as manifested in Recommendation 2006/962/EC, dated 18 December 2006 [12] and 22 May 2018 [13] by the European Parliament and the Council of the European Union. Both regard them as one of the Fundamental Competencies for citizens. The European Commission [10] incorporates a set of descriptors such as, information and data literacy, communication and collaboration, media literacy, digital content creation (including programming), security (encompassing digital well-being and cybersecurity-related skills), matters pertaining to intellectual property, problem-solving, and critical thinking [11].

In this context, a digital competence reference framework has been established. The Joint Research Centre (JRC) of the European Union introduced the "Digital Competence: Identification and European-wide validation of its key components for all levels of learners (DigComp) project in 2010. In its most recent rendition, DigComp 2.2, it retains the same five competency areas as its preceding version 2.1: information and data search and management; communication and collaboration; digital content creation; security; and problem-solving. What the 2.2 update contributes are examples of knowledge, skills, and attitudes pertinent to each competency [35]

1.2 The Spanish national plan for digital competencies

This study is situated within the domain of information and media literacy as advocated by DigComp 2.2, an action framework that is integral to the Spanish National Plan for Digital Competencies set forth by the Government of Spain [18]. Among the examples proposed in version 2.2 are Misinformation and disinformation on social networks and news sites related to information and media literacy." This challenge is encompassed within competency area 1, denoting information and media literacy. Training in literacy competence encompasses, for instance, "the capabilities to discern and utilise various types of sources, to search, gather, and process information." Such skills become imperative when assessing online content and its origins, a proficiency that is an essential facet of information literacy.

To ensure the rights of individuals in their interactions with new technologies, the Government of Spain [19] has adopted the Digital Rights Charter with a pronounced inclusive perspective, aiming to mitigate the digital divide. The objective of fortifying the development of critical thinking that aids in distinguishing objective facts from mere unsubstantiated opinions, as well as identifying fake news and disinformation, is articulated in section XVII of right 3, titled "Participation and shaping of the public sphere.

1.3 Gamification and sustainability

Gamification in education refers to the use of game elements to engage and motivate students during the learning process. It is seen as a tool to improve student skills, maximize learning, and increase motivation [6]. Gamification, understood as the application of mechanics and playful elements in non-playful contexts, has proven to be a versatile and effective methodology beyond the educational environment. López et al. [21] highlights how gamification can be used in museums to connect with users and influence their behavior. Carpintero [7] presents a case study of gamification in primary education, showing how it enhances student motivation and active participation in the learning process.

In the Spanish educational context, gamification has garnered considerable interest, as evidenced by various research studies and educational practices. The research conducted by González Calatayud [20] explores the implementation of Escape Rooms in Vocational Training (FP). This pedagogical innovation, grounded in the concept of Gamification and Game-Based Learning (ABJ), is notable for its potential to enhance motivation and learning, problem-solving, and challenges, and reinforces not only specific knowledge but also skills such as critical thinking and problem-solving.

The study by Galiç and Yıldız [15] highlights that while gamification in mathematics classes does not show a significant increase in academic performance, it does promote notable improvements in students' motivation and attitude towards the subject. García Martínez and Fuentes Agustí [16] point out an upward trend in the adoption of playful strategies assisted by ICT in education, emphasizing how these practices foster a positive change in students' attitudes and motivation towards learning.

In line with our pretest and posttest design, it is relevant to mention the study by Morón Hernández et al. [24], which examines gamification as a strategy for hybrid education in university students. This study, employing a similar approach, reported significant improvements in students' learning autonomy following the implementation of gamification. Specifically, the authors observed that while in the pretest only 42.2% of the students achieved a regular grade level, in the posttest, this figure rose to an impressive 94.7% with good grades, highlighting the positive impact of gamification in hybrid education.

This study aligns with the 2030 Agenda and the Sustainable Development Goals (SDG) (UN, n.d.) [34], specifically SDG 4: Quality Education and SDG 13: Climate Action. Moreover, the ethical framework is associated with the Earth Charter, striving for the construction of a more just, sustainable, and peaceful global society through value-based education, thereby creating spaces for reflection to facilitate better decision-making [9]. It is imperative to safeguard our planet's health [3] by embedding sustainability in educational systems, so that students grasp the importance of our world and act for its preservation. The European Commission has prioritised environmental education in the forthcoming years, materialising in the establishment of the European competency framework for sustainability, GreenComp, which is an integral component of the European Green Deal [26].

In relation to SDG 4, the study proposes introducing gamification into the classroom with an aim to motivate students to design an interactive game that enables discernment between accurate information and disinformation. Various authors have utilized gamification methodology to address emotions such as empathy [17], counteract fake news through simulation techniques [27], and adopt roles from the perspective of a disinformation agent [23]. In this research, a pre-designed video game will not be used; instead, students will be provided with the knowledge to create an adventure game wherein they must navigate the daily flood of climate change misinformation on the Internet.

In connection with SDG 13, the GreenComp framework is referenced, which aims to cultivate a sustainability mindset that encourages the development of competencies to act empathetically, responsibly, and caringly for our planet. Sustainability is a multifaceted concept that entails acquiring an array of skills, such as systemic thinking, critical analysis, contextualising issues, risk assessment, and strategic planning. Grasping this environmental crisis, responding effectively, and collectively addressing it are essential facets in the quest for solutions. The cultivation of green competencies based on knowledge and attitudes can foster responsible action, as well as ignite the drive to adopt or advocate for measures at local, national, and global scales. Acquiring sustainability-related skills can assist students in overcoming cognitive dissonance arising from awareness of a situation but lacking the capability to act upon it.

GreenComp encompasses four competency areas associated with understanding sustainability. Table 1 presents the descriptors and dimensions corresponding to the Second Area, which is the focus of this study:

2 Materials and methods

Considering the aforementioned theoretical framework and the limited empirical studies demonstrating the impact of gamification on the development of critical thinking in the face of disinformation, a study is designed to assess the influence of a training programme that incorporates the creation of an educational Escape Room among undergraduate students.

Table 1 Area 2 of competencies and descriptors from greencomp

Area	Competency	Descriptor
2. Embracing the Complexity of Sustainability	2.1. Systems thinking	Approach a sustainability issue from all angles; consider time, space, and context to understand how elements interact within and between systems
	2.2. Critical thinking	Evaluate information and arguments, identify assumptions, challenge the status quo, and reflect on how personal, social, and cultural contexts influence thinking and conclusions
	2.3. Problem contextualization	Frame current or potential challenges as a sustainability problem in terms of difficulty, stakeholders involved, time, and geographical scope, with the aim of identifying suitable approaches to anticipate and prevent problems, as well as to mitigate existing ones and adapt to them

Source: [11]

Presently, a multitude of resources exist for generating interactive multimedia projects that offer online gaming opportunities. This research employs Genially, an application extensively vetted in educational settings, to produce interactive 2.0 content. For this study, a predefined template from the category Gamification > Escape Room titled 'Breakout Save the Planet' was selected.

2.1 Objectives and research questions

Based on the presented theoretical framework, the general objective of this research is to assess whether the role of the student as an educational software developer offers an opportunity to promote critical consumption of online content, aiming to contribute to the fight against disinformation.

The research questions that guide this study are as follows:

- Might educators utilize gamification as a pedagogical approach to confront the challenges posed by the digital divide and misinformation?
- Does enhancing digital skills among students to become developers of educational software encourage critical content consumption?

The initial hypothesis posits that the implementation of a technology training programme will significantly contribute to the acquisition of skills for identifying falsehoods and deceptive information on the internet.

2.2 Variables

In this research, we engage in an in-depth examination of two principal categories of variables: independent and dependent, alongside polytomous nominal variables.

Independent Variable: Our independent variable is the employment of a virtual Escape Room to classify and collate false news with true news. This gamified tool is deployed to assess its impact on students' critical abilities in identifying and debunking false information.

Dependent Variables: The dependent variables in this research are the dimensions of critical capacity achieved by students in debunking misinformation. These dimensions are evaluated through a question with four answer options, reflecting different reactions to news on social media:

- (a) *Decontextualization* Assesses whether the student suspects that the images in a video have been taken out of context.
- (b) *Practical Reasoning* Measures the student's ability to apply common sense in inconsistent situations.
- (c) *Source Verification* Determines whether the student seeks to verify the authenticity of information through alternative sources.
- (d) *Fact-checking* Evaluates the student's proficiency in discerning manipulations in images and verifying information through advanced searches.

Polytomous Nominal Variable:

- (a) *Educational Level* For the purposes of our study, we have focused on the category of "university" in its third year of undergraduate study. This classification allows for an analysis of how the university level specifically influences students' critical capacity to identify and debunk fake news.

2.3 Research context and participants

The research is conducted using an experimental design with pretest and post-tests measures. The sample for our study comprised 107 subjects, all of whom were students at the Universidad CES Don Bosco, a private institution affiliated with the Universidad Complutense de Madrid. These participants, aged between 19 and 21 years, represent a homogeneous cohort

Table 2 Research stages

Stages	Actions	Variable
Pretest	20 questions	Dependent
Training program	Design of escape room	Independent
Posttest	1 question extracted from the pretest	Dependent

Table 3 Training Program (Intervention)

Session	Contents	Compe- tency area	Competence	Skill level
1	Advanced Google searches	1	1	Basic
	Locating news related to climate change	1	3	Basic
2	Evaluating the information found. Online resource Fact Checker to detect hoaxes and misinformation in any field. PANTERA method for detecting misinformation (origin, authorship, novelty, tone, evidence, reply, expand)	1	2	Basic
3	Gamification with genially. explanation of the “Breakout save the planet” template	3	10.11	Intermediate
4	Development of the Escape Room. A minimum of 4 missions is required with the following stipulations: The 4 missions must have a narrative thread referring to climate change. Each news explanation must have an argument based on the PANTERA method	5	20	Highly specialized

in terms of age, providing a consistent foundation for analysis. Each student is enrolled in the same Information and Communication Technology course, uniquely positioning them for future professional roles in the educational sector.

2.4 Procedure

The three phases of the research are: measurement of the dependent variable (pretest), followed by the implementation of the training program (intervention), and finally, the assessment test is repeated (Posttest) as described in Table 2.

The intervention entailed the execution of four working sessions. Each session spanned approximately two hours per day. Throughout the course of the activities, students collaborated in small groups (comprising 4–5 members). The sessions were structured in alignment with the predetermined objectives. Table 3 summarizes the intervention program:

2.5 Data collection instrument and data analysis

The level of students’ digital literacy concerning critical content consumption (pretest) was analysed using the questionnaire published by the Exit Foundation [14]. This instrument evaluates the digital verification processes young individuals employ when consuming online content. It is part of a training and digital innovation program, involving partners such as the Sabadell Bank Foundation, Mobile Week (from Mobile World Capital Barcelona), and Learn to Check as the educational partner. The instrument consists of 20 items, each with four response options, of which only one is correct. A point (1) is awarded for each correct answer and zero points (0) for each incorrect response.

For the post-test segment, wherein we measure students’ acquisition of skills to identify online falsehoods and deceptive information, the same questionnaire is used, but only the selected question posing the greatest challenge is considered.

After coding the collected data, analysis was carried out using the SPSS software, version 27. An initial descriptive analysis (frequency tables) was conducted on the data. Subsequently, relationships between the variables were examined through contingency tables, employing a non-parametric statistical analysis via a Chi-square test.

3 Results

3.1 Descriptive results

Figure 1 presents a comparative analysis between the three student groups, showcasing the average scores for each of the 20 questions. The results suggest there is an urgent need to advocate for media and digital literacy among digital natives. While these individuals have developed an intuitive understanding of the web's interconnectedness, they often lack structured education that enables them to identify the potential risks inherent in it, such as misinformation, data theft, identity fraud, and phishing. Concerning the most challenging responses, we find that they relate to critical online content consumption (7), digital footprint (12), advanced search techniques (16), passwords and security (18), and copyright (19).

The data gleaned from both the pretest and post-tests in this study place particular emphasis on question 7, as depicted in Fig. 1 and detailed in Table 4.

In the pretest, responses spanned all four categories (A, B, C, and D). Notably, C accounted for 64.4% and D for 25.4%. The post-tests results showed responses only in the categories C (76.3%) and D (23.7%). It's worth highlighting that the majority of the post-tests responses were in category C. Furthermore, 54.2% of the participants chose option C in both the pretest and post-tests. Detailed data can be observed in Table 5, respectively.

In the bar chart depicted in Fig. 2, it is evident that those who responded with "A" in the pre-assessment and "C" in the post-assessment account for (5.08%) of the total. It is noteworthy that those who answered "C" in both the pre and post assessments represent (54.2%) of the total. These specifics can be viewed in Fig. 2.

Table 6 aims to compare the two categories that appear in both the pre and post assessments, namely "C" and "D". Chi-square tests, with the McNemar alternative, will be applied to these data.

The categories that did not appear in both the pre and post assessments have been filtered out to discern any significant differences in the variable "If you see a video on YouTube where it's said that immigrants coming to Spain are throwing food into the sea". It is notable that those who responded with "C" in the pre-assessment represent 71.7%, while in the post-assessment, they account for 77.4%. In contrast, those who answered "D" in the pre-assessment account for 28.3%, and in the post-assessment, 22.6%. We can also observe that those who responded with "C" in both the pre and post assessments represent 60.4%

Subsequent to the initial analysis, McNemar chi-square tests will be employed to ascertain the presence of statistically significant variances between the pre-assessment and post-assessment outcomes, as delineated in Table 7.

In the test, it can be noted that the significance level is 0.607, which is greater than 0.05. Therefore, no significant differences are observed in this variable between the pre and post responses.

Given that the value is greater than 0.05, the null hypothesis is upheld.

Fig. 1 Average scores of the three groups in the pretest questionnaire

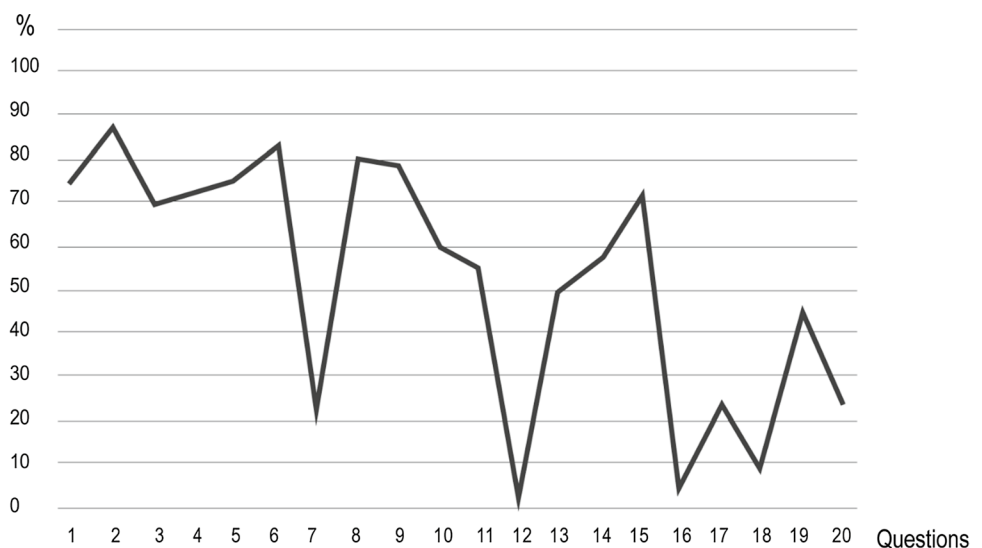


Table 4 Question 7 and its dimensions A, B, C, D for the dependent variable: critical capacity achieved by students to debunk a hoax

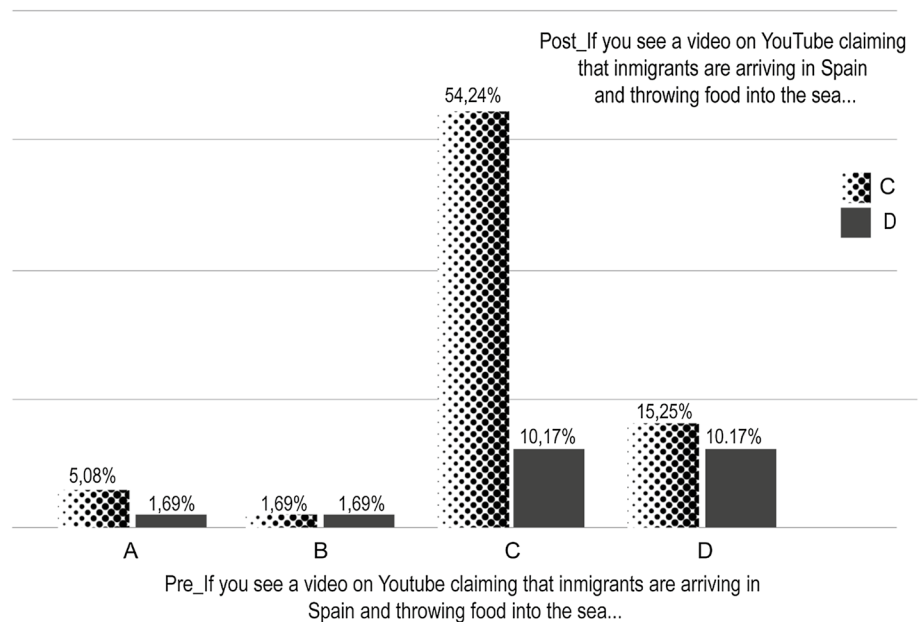
Pre and post	Possible responses (A, B, C, D)	Categories	Explanation
If you see a video on YouTube that claims immigrants are arriving in Spain and throwing food into the sea...	(A) It seems suspicious to me that the food boxes have texts in Arabic and French and not in Spanish	Intuition	I suspect the images shown in the video have been taken out of context
	(B) Come on! How could that be true? Common sense tells me that this makes no sense at all	Practical reasoning	I apply common sense to an incoherent situation
	(C) I conduct a search on YouTube using keywords to see if there are similar videos and verify whether the video in question is manipulated or taken out of context	Source verification	I check the veracity in alternative sources by conducting advanced searches
	(D) All of the above answers are correct	Fact verification	All checks are verified: the images have been manipulated, the content lacks common sense, and I verify its authenticity through advanced searches

Table 5 Frequency table and percentages of the analyzed variables

Pre_If you see a video on YouTube claiming that immigrants are arriving in Spain and throwing food into the sea * Post_If you see a video on YouTube claiming that immigrants are arriving in Spain and throwing food into the sea

% of Total

		Post_ If you see a video on YouTube claiming that immigrants are arriving in Spain and throwing food into the sea		Total %
		C %	D %	
If you see a video on YouTube claiming that immigrants are arriving in Spain and throwing food into the sea	A	5.1	1.7	6.8
	B	1.7	1.7	3.4
	C	54.2	10.2	64.4
	D	15.3	10.2	25.4
Total		76.3	23.7	100.0

Fig. 2 Pre and post graph

3.2 Resultados empíricos

Regarding the final Escape Room work, the following Table 8 analyses how competency 2.2, the focus of this study, has been evaluated.

Table 6 Significant differences in the study variable

% of Total	Post_ If you see a video on YouTube stating that immigrants are arriving in Spain and discarding food into the sea			
		C %	D %	Total %
Pre_If you see a video on YouTube stating that immigrants are arriving in Spain and discarding food into the sea	C	60.4	11.3	71.7
	D	17.0	11.3	28.3
Total		77.4	22.6	100.0

Table 7 Chi-square tests

	Value	ExactSig. (2-sided)
McNemarTest		0.607 ^a
N of valid cases	53	

^aBinomial distribution used

4 Discussion

Through the exploration of the impact of gamification on the development of critical thinking in the face of disinformation, this study has contributed new findings to research on how the role of students as educational software developers allows the scaling of digital competency levels set by DigiComp. Focusing on Competency Area 1: Search and Management of Information and Data, the study addresses competency 1.2: Evaluate Digital Data, Information, and Content. Prior to the training program, students exhibit a basic competency level, being able to discern the reliability and integrity of common sources of digital data, information, and content. Upon completing the training program, as indicated by post-intervention results with a variance of 5.7% (Pre C 71.7%, and Post 77.4%), students achieve an advanced competency level. Milestones achieved include the ability to conduct an evaluation of the reliability and integrity of various information sources, data, and digital content, and to critically assess the trustworthiness and integrity of such sources.

Regarding Competency Area 3: Creation of Digital Content, competencies 3.1 (content development: create and edit digital content in different formats, express oneself through digital media) and 3.2 (integration and reworking of digital content: modify, refine, enhance, and integrate information and content into an existing body of knowledge to create new, original, and relevant content) were addressed. Students begin at an intermediate level and, by the end, achieve a highly specialized level, creating solutions to complex problems with limited definition related to content creation and editing in various formats, and self-expression through digital media.

The questionnaire results suggest that students are aware of the necessity to pinpoint the origin of the information gleaned from the internet (e.g., on social media platforms) and to validate it using multiple sources. This is with the intent to recognize and understand the viewpoint or bias underpinning particular pieces of information and the sources of data. However, the absence of responses A and B in the post-assessment indicates a deficiency in intuitive and practical reasoning.

It is essential to underscore, based on the outcomes, that the training program imparts on students an awareness of the significance of identifying and verifying the origin of information obtained from the internet, primarily on social media platforms. Nevertheless, post-assessment results point to a shortage in the students' problem-solving

Table 8 Domain 2 of competencies and descriptors of greencomp

Domain	Competence	Descriptor
2. Embracing the complexity of sustainability	2.2 .Critical thinking	Knowledge: Is aware that sustainability claims without solid evidence are often mere communication strategies, also known as greenwashing; Skills: Can analyze and evaluate arguments, ideas, actions, and assumptions to ascertain if they align with evidence and values concerning sustainability; Attitudes: Trusts in science even when lacking some of the necessary knowledge to understand scientific claims

skills, evident from the absence of responses for items A and B and a decrease in Post with D responses from the Pre-assessment (28.3%) to the Post-assessment (22.6%).

Studies on the ramifications of disinformation have underscored the urgency to formulate effective strategies to combat the spread of fake news [31]. Some of these tactics hinge on the deployment of technology aimed at curbing the disinformation effect and bolstering the critical assessment of information [28]. Simultaneously, there's a need to preserve the authority of science as a legitimate institution that has accrued power without political justification [5].

The development of new strategies to prevent the dissemination of misinformation and the conditions under which these strategies may be successful are consistently highlighted in current research [4]. Preventative strategies such as inoculation [2] and prebunking [27] are often utilized. Inoculation prepares the audience to detect misinformation by providing weakened examples of how they might be misled. Prebunking, on the other hand, teaches techniques to highlight facts that discredit false information before its spread. Studies have shown that prevention (prebunking) is more effective than subsequent debunking.

The present research was designed to explore the efficacy of new strategies for preventing the spread of misinformation and to develop verification techniques (prebunking) with the aim of reducing the risk of future education professionals inadvertently becoming mouthpieces for hoaxes. Although the chi-squared test significance result of 0.607 does not support our hypothesis, other studies focusing on the student's role as a game user have shown positive outcomes.

A recent study by Quevedo-Redondo [27] assessed the use of newsgames, games with journalistic purposes, in a higher education environment (Journalism undergraduate students). It aimed to address the risks associated with excessive exposure to harmful content via social media platforms. The experiment was conducted using The Bad News Game, developed by the Dutch association of journalists and academics, DROG. This tool allowed students to acquire protective skills and prebunking strategies through gamification, building on the inoculation theory [22] proposed by Quevedo-Redondo and others [27].

The study by Morejón-Llamas [23] focused on the burgeoning phenomenon of fake news during the pandemic, considering the extensive use of social media by young individuals. The video game Go Viral! was assessed as a tool to broach the topic of misinformation in both pandemic and post-pandemic settings. The results showed that the game, grounded in psychological inoculation, enabled users to understand the process of generating misinformation online, its amplification by echo chambers, and the necessary tools to establish online communities from the perspective of a disinformation agent, offering data transmission, contextualization, sensitization, interaction, and stimulation.

Manso Rodríguez conducted a descriptive investigation aiming to apply gamification in libraries to promote media and information literacy. Through literature analysis, it has been demonstrated that gamification is an effective tool to motivate library users to seek ongoing training in media and information literacy [29].

5 Conclusions

This research focuses on competency 2.2 Critical Thinking from the domain "Tackling the Complexity of Sustainability" as set out by DigiComp. It is defined as a cognitive process that allows for the evaluation and understanding of information regarding sustainability issues, expanding horizons without undermining the information and its sources. Critical thinking helps students challenge and change their values, perspectives, and understanding of the world, as well as scrutinise prevailing norms, practices, and opinions to consider their own values, perceptions, and actions and take an informed stance in relation to sustainability discourse [25, 30].

The aim of this work, related to the research questions, focuses on assessing whether the role of the student as an educational software developer provides an opportunity to foster their critical thinking, and thereby their critical online content consumption, hence aiding the fight against misinformation.

The initial hypothesis suggests that the implementation of a technology training program would significantly enhance the acquisition of skills for identifying hoaxes and falsehoods online. Although the results did not meet the established statistical threshold, it is proposed as a potential research avenue to conduct future studies designing training programs with a more extensive digital training curriculum centred on other methodologies, evaluating their effectiveness in preventing online hoaxes and falsehoods.

In the practical objective, students are urged to use the PANTERA method (source, authorship, novelty, tone, evidence, replicas, expand) to reason about the climate change hoax. In 95% of the works, the 'source' and 'authorship' dimensions are applied, but the rest of the items, related to intuition and practical reasoning categories (A and B) answered in the Pre (5.08% and 1.7%), were not retained in the Post, which responded to the 'source verification' category. The

'fact verification' category decreases in the Post, moving from 28.3% to 22.6%. This reduction could be attributed to the understanding of both dimensions as mutually exclusive.

Students were expected to answer 'D' to the presented question since this response encompassed the three dimensions: intuition, practical reasoning, and source verification. A dystopian question about a video in which a group of immigrants arrive in Spain and throw food into the sea was proposed. Students were expected to distrust the veracity of such material (A and B) and to confirm their suspicions, resort to advanced searches (C), as taught through training plans.

A total of 107 students participated in this study, consisting of 64 women and 43 men, within the framework of the Information Technology subject. Participants displayed a high level of competence in using advanced operators to retrieve information and skill in quickly debunking news. However, the need was identified to provide examples and personal uses to reinforce this skill as they abandoned logic and intuition as the first filter when faced with a hoax.

Throughout the subject, extensive debates were conducted to find ways to debunk inappropriate information. Hoaxes are found both online and in everyday life. The presence of sensationalist content and attention-grabbing calls by issuers leads to situations where we are left bewildered. In these cases, a way has been sought not to enter into conflict but, at the same time, maintain a secure but doubting attitude towards the misinformation messenger.

Promoting critical thinking is a shared responsibility across all subjects. Students who think critically are able to examine their views, question claims or assumptions, and demonstrate intellectual empathy. This contributes to ongoing growth and development as they can dynamically consider diverse viewpoints over time.

Regarding gamification, there's a noticeable motivation and performance in gamification designs, as well as satisfactory results from news selection and reasoned criteria for debunking information. However, the selected tool does not allow recording user hits, so results could not be provided in this regard.

This preliminary study has several limitations to consider. Having such a small sample size and 100% of the sample from the technology subject makes it hard to generalise results. From these outcomes, different research avenues emerge that can evaluate the result of educational action in the fight against misinformation. In future work, more implicit cognitive aspects should be considered, generating a greater number of descriptors that allow us to delve deeper into the dimension of critical thinking.

Author contribution The author declares that she has contributed equally in the following sections: Made substantial contributions to conception and design, or acquisition of data, or analysis and interpretation of data; and Been involved in drafting the manuscript or revising it critically for important intellectual content; and Given final approval of the version to be published. The author should have participated sufficiently in the work to take public responsibility for appropriate portions of the content; and Agreed to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved.

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Data availability The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Conflict of interest The author declares that there are no conflicts of interest with respect to the research, implementation, and publication of this article.

Ethical approval This project is carried out within the teaching activity of the Department of Artistic, Bodily, and Musical Expression (Plastic and Visual Arts) of the CES Don Bosco University of Madrid and therefore has the approval of the Research Ethics Committee of the CES Don Bosco University of Madrid.

Informed consent The participants have been informed and have signed an informed consent document. The procedures used in this study adhere to the tenets of the Declaration of Helsinki.

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